Simulating Neutronic Behaviour and Tritium Breeding in Heterogeneous Fusion Breeder Blankets

MPhys Research Project

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Abstract

Breeder blankets composed of lithium compounds are used to produce tritium required to power fusion reactors. In semester one we used a dedicated Monte Carlo framework to model the neutronics and calculate a Tritium Breeding Ratio (TBR). At the outset of the second semester several modifications were made to broaden the scope of our investigation and improve our understanding of breeder behaviour. Chief among these was the adoption of a heterogeneous, layered blanket configuration. The aim of our investigation was to study the neutronic response of a water cooled $Li_{17}Pb_{83}$ breeder at various geometric configurations. Measurements of the Tritium Breeding Ratio (TBR) were the primary metric used to evaluate breeder performance. Other parameters such as neutron multiplication and material damage were also used to characterise the blanket behaviour. Identifying the optimum blanket geometry to maximise tritium breeding remained a complex and time consuming challenge. A Bayesian optimisation machine learning programme was identified as the most efficient method to achieve this. An unconstrained 8-dimensional machine learning workflow produced a peak TBR of 1.4055$\pm$0.0033 while maximising the breeder volume and minimising the coolant. The algorithm identified the foremost layers as the most active contributors to the total TBR.

1 Introduction

Deuterium-Tritium (D-T) fusion has long been considered to be the optimal reaction for use in a commercial fusion reactor (Heckrotte and Hiskes, 1971).

$$\mathrm{D} + \mathrm{T} \longrightarrow \alpha + \mathrm{n} + 17.6\,\mathrm{MeV} \tag{1}$$

Its large reaction cross section and energy output are key factors in its selection. Unfortunately, it has a key downside. Tritium is a radioactive isotope with a half-life of 12.32 years (NNDC, 2024), so it only exists in trace amounts on Earth. Consequently, as discussed in previous semester's report (Peters, 2025), there is a significant body of work focused on the reliable breeding of tritium for use in fusion reactors. Instead of relying on external tritium production, many reactor designs rely on a breeder blanket. The reactor vessel is surrounded with lithium which when bombarded with fusion neutrons releases tritium as shown in the equations below.

$$^{6}\mathrm{Li} + \mathrm{n} \longrightarrow \alpha + \mathrm{t} + 4.8\,\mathrm{MeV} \tag{2}$$

$$^{7}\mathrm{Li} + \mathrm{n} \longrightarrow \alpha + \mathrm{t} + \mathrm{n} - 2.5\,\mathrm{MeV} \tag{3}$$

The primary reaction involved in tritium breeding involves ^{6}Li and consumes one neutron to produce one triton. Clearly this one-to-one neutron economy is not sustainable if any parasitic absorption takes place. Thus, reliable tritium production greater than the threshold of equilibrium requires the production of additional neutrons. Elements such as lead undergo a multiplication (n,2n) reaction upon interaction with neutrons and so are mixed into the breeder material.

$$^{208}\mathrm{Pb} + \mathrm{n} \longrightarrow {}^{207}\mathrm{Pb} + 2\,\mathrm{n} - 7.4\,\mathrm{MeV} \tag{4}$$

With many prototype reactors planned for construction by the 2040's (Barbarino, 2018) there is significant interest in developing and refining the principles of tritium breeding. Breeder structures are evaluated in terms of their Tritium Breeding Ratio (TBR) using Monte Carlo simulations of neutron transport. A TBR of greater than 1.05 is typically deemed necessary to maintain equilibrium (Tanabe, 2017). In the first half of our project (Peters, 2025) we performed neutronics simulations on a model blanket using OpenMC, a dedicated neutron transport Monte Carlo simulation (Romano et al., 2015). A water cooled blanket consisting of a eutectic lithium-lead alloy ($Li_{17}Pb_{83}$) produced the best TBR results. However, our investigation had significant limitations. Unlike practical blanket designs which separate the breeder and coolant into a series of distinct, alternating layers (Lee et al., 2012; Qu et al., 2020; Boullon, Jaboulay and Aubert, 2021) our simulation used a homogeneous mixture of the two. Additionally, we used a simplified ring neutron source instead of a more realistic plasma body source. By utilising more complex blanket designs we hope to gain a deeper understanding of their performance requirements. It is necessary to carry out investigations beyond simple measurements of TBR and to consider other factors that may favour or rule out certain configurations. Parameters such as multiplication rate, neutron flux and damage energy can have significant implications for blanket performance and operational lifetime.

Another flaw in our implementation during the homogeneous blanket studies was a lack of multidimensional investigation. Simulations were carried out by varying individual parameters making it difficult to isolate the primary influences on the behaviour of TBR. This is made even more difficult and time consuming as heterogeneous effects are introduced. A Machine Learning (ML) programme is beneficial in this regard (Zhou, 2021), allowing for the investigation of complex, multidimensional variables.

2 Theory

2.1 Heterogeneity in Blanket Designs

Heterogeneous neutron transport simulations can exhibit radically different behaviour to simpler, homogeneous models. For example, a study of fast fission reactors by Ibrahim et al. (2021) demonstrates that while homogeneous simulations are much faster and easier to model they can fail to exhibit the same power distribution and fuel transmutation rate. It is reasonable to assume that the neutron economy of a fusion reactor will also be influenced by heterogeneity. Investigations by Lee et al. (2012) and Qu et al. (2020) suggest that a heterogeneous breeder blanket would have a reduced TBR compared to a homogeneous simulation.

2.2 Material Damage

In addition to evaluating the TBR of heterogeneous blanket structures we also recorded several other tallies including the damage inflicted on the structural components. When irradiated with neutrons the atoms in a material are displaced from their position in the lattice. The number of displacements (N_d) depends on the damage energy delivered to the material (T_d) and the threshold displacement energy (E_d),

$$N_d = \frac{0.8T_d}{2E_d}\left((1-c)\left(\frac{0.8T_d}{2E_d}\right)^b + c\right) \tag{5}$$

where b and c are material constants that express the recombination rate of displaced atoms (Saha, Devan and Ganesan, 2019). Extrapolating this value to full breeder structures requires conversion to displacements per atom (dpa). In a reactor operating at power P the dpa per full-power-year (fpy) can be expressed as

$$\mathrm{dpa\,fpy}^{-1} = \frac{N_d}{\left(\frac{\rho V N_A}{\mu}\right)} \cdot \frac{P}{E_f} \cdot (365.25 \cdot 24 \cdot 60^2) \tag{6}$$

for a structure of volume V, density ρ and molar mass μ. The energy released during fusion is represented by E_f.

High-energy neutron flux has a wide variety of effects on irradiated materials. A critical one for fusion materials is helium induced embrittlement. Helium produced in transmutation reactions accumulates at grain boundaries, causing swelling and weakening the structure (Trinkaus, 1985). It is possible to express critical helium concentrations as a lifetime dpa limit for selected materials (Gilbert et al., 2012).

2.3 Machine Learning

The tritium breeding behaviour of heterogeneous breeder blankets is a complex and multivariate problem. While it is possible to isolate trends by manually investigating a wide range of blanket configurations this process is incredibly time consuming and it is easy to miss important results. Instead, many neutronic simulations rely on Machine Learning (ML) algorithms to optimise tritium breeding (Barker, Gilbert and Edmondson, 2026; Humphrey et al., 2026). Bayesian optimisation is the standard method applied in these circumstances (Shahriari et al., 2016). Essentially, an initial guess is updated with new data to produce a better prediction.

Prior to the optimisation process a set of datapoints is generated at random positions in the model space. These values are used to produce a surrogate model, a function that aims to mimic the behaviour of the simulation. Instead of modelling $\sim 10^6$ neutrons for every possible datapoint a surrogate is defined based on the mean (μ) and variance (σ) of each datapoint in the set. Gaussian Process Regression (GPR) is a popular surrogate in neutronics modelling since it requires a relatively small sample size and can easily account for stochastic noise resulting from the probabilistic nature of neutron transport (Polke, Ahle and Söfker, 2026). Once a surrogate has been created it must be optimised. This process is driven by an acquisition function; a mechanism that selects where in the model space to carry out the next investigation. The Upper Confidence Bound (UCB) method makes the selection based on the mean and variance of the surrogate function (Mia et al., 2025; Ribeiro et al., 2026).

$$\alpha = \mu(x) + \sqrt{\beta}\sigma(x) \tag{7}$$

Higher values of β prioritise the investigation of areas with high uncertainties. The process of testing, acquisition and simulation is repeated until the surrogate model converges on the optimised value.

3 Methodology

3.1 Blanket Structure

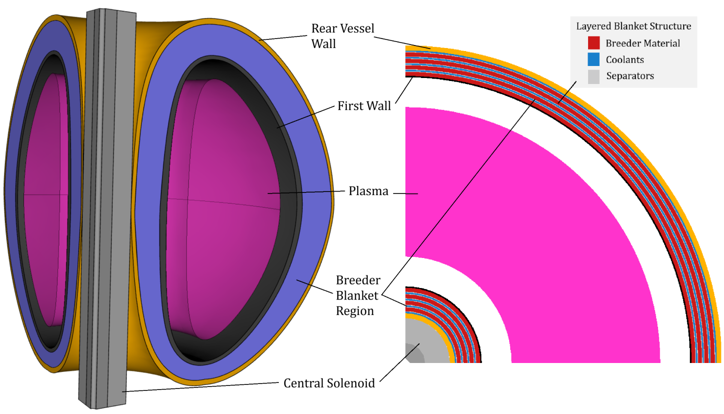
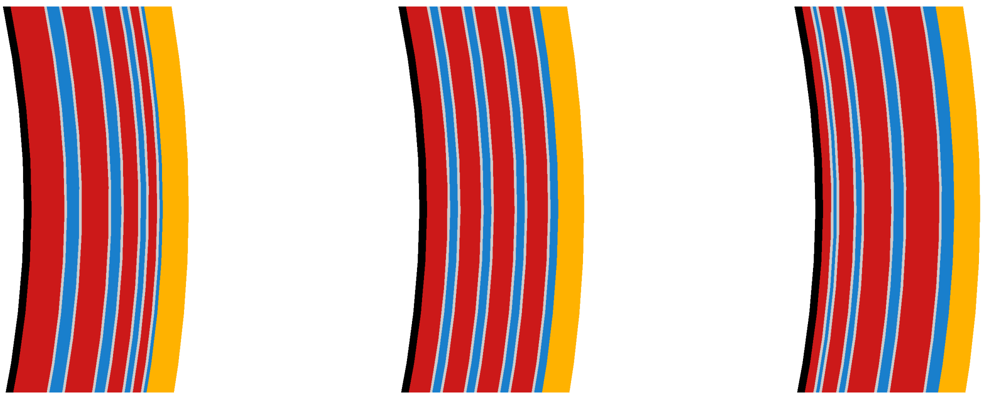


Figure 1: A pair of diagrams depicting the breeder dimensions and layered configuration. On the left, a 3D model shows the shape and arrangement of the overall structure. On the right a 2D slice demonstrates the layered formation of the heterogeneous blanket.

Many heterogeneous breeder blankets are designed with several ring-like layers, with alternating breeder material and coolant separated by structural material. There is significant variability in design, ranging from only two layers (Gohar et al., 1998; Segantin et al., 2020) to five (Lee et al., 2012; Qu et al., 2020). A four layer configuration (as demonstrated in Fradera et al., 2021) was selected for this investigation. As depicted in figure 1, the blanket is shielded by a 25 mm thick first wall consisting of Zircaloy-4 (King et al., 2022).

The available literature supports the placement of the breeder as the next layer before the coolant (Klix et al., 2005; Boullon, Jaboulay and Aubert, 2021). A eutectic $Li_{17}Pb_{83}$ breeder and water coolant were selected with thicknesses of 80 mm and 30 mm respectively. The lithium in the breeder layers was enriched to 40% ^{6}Li. Structural layers consisting of low-activation Eurofer97 steel (Stornelli et al., 2024) separated each breeder and coolant. The material composition of each blanket segment was chosen for its neutronic properties demonstrated in the homogeneous blanket simulations (Peters, 2025).

To comprehensively investigate the effects of heterogeneity on TBR it was important that we attempt to explore the effects of modifying individual layer geometry. Unfortunately, it would be too time consuming to simulate each individually. Instead, a factor denoted by ζ was used to impose a linear gradient on the thicknesses of each layer. Each breeder and coolant pair had their thickness altered by a factor of $1-2\zeta$, $1-\zeta$, $1+\zeta$ and $1+2\zeta$ sequentially, as shown in figure 2. Thus, negative values of ζ led to thicker front layers (figure 2a) and positive values led to thicker rear layers (figure 2c). Implementing this gradient meant that the relative impact of breeder location could be studied more effectively. It is important to note that thicknesses of the first wall and separators were kept constant throughout.



(a) Front-loaded (ζ=-0.3) (b) Standard (ζ=0) (c) Rear-loaded (ζ=0.3)

Figure 2: 2D cross sectional diagrams of the breeder blanket layers, beginning with the first wall (black) and then alternating between breeder (red) and coolant (blue) layers separated by structural layers (grey). The rear containing wall is highlighted in yellow. Each slice shows the relative thicknesses of the layers for values of ζ.

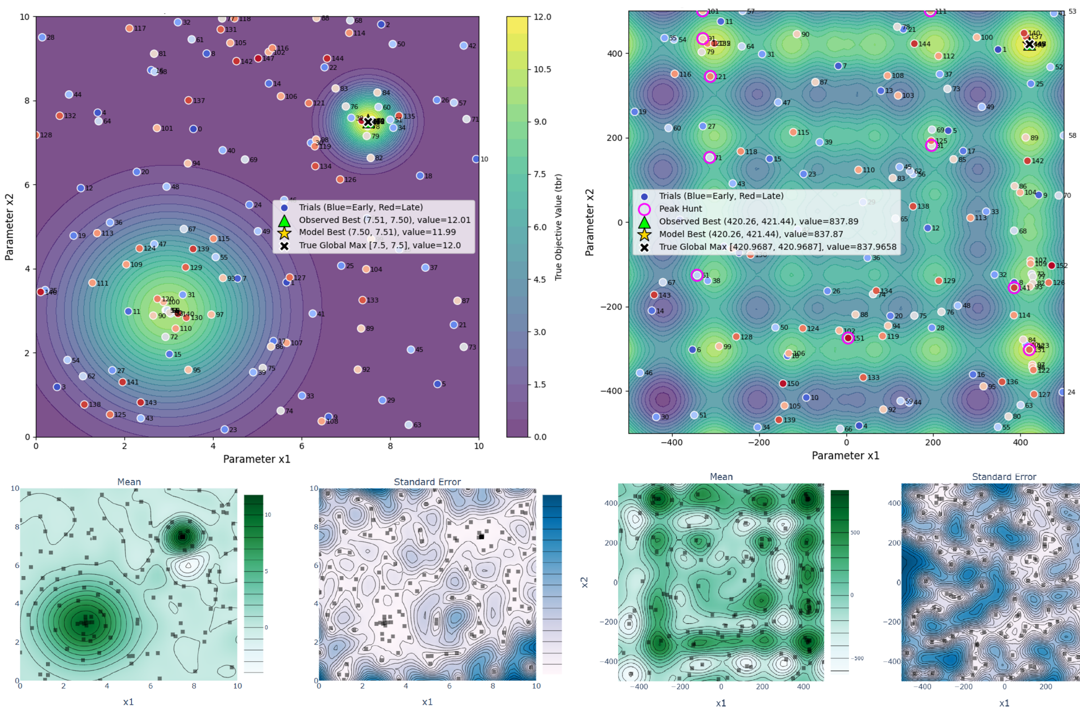
3.2 Alternative Simulation Tallies

The tally prioritised in a majority of breeder simulations is TBR. It provides a simple method of comparison, focusing on the main goal of breeder blankets which is the production of tritium. A value of $\geq$1.05 is typically selected as a threshold for sustainable tritium breeding. However, there are many other factors that influence the selection of breeder materials and dimensions. OpenMC provides several tallies that can be used to probe the full functionality of the blanket and investigate the neutronics in more detail (Romano et al., 2015).

To achieve a TBR greater than equilibrium it is vital that sufficient neutron multiplication reactions take place. The OpenMC 'multiplication' tally counts the number of $(n, 2n)$ reactions that take place per source neutron. We would expect a vast majority of these reactions to take place within the breeder layers containing the lead multiplier. When neutrons interact with nuclei they are either absorbed or scattered. The rate at which these reactions take place per source neutron is measured by the 'absorption' and 'scattering' tally respectively. In a simulation with minimal leakage we would expect the total absorption to be equal to the one plus the multiplication value. Since neutrons can be scattered several times before they are absorbed the 'scattering' tally should record much higher values. During these interactions neutrons also impart energy into the materials, measured in eV per source neutron by the 'heating' tally. The energy that results in atomic displacement from the crystal lattice is logged by 'damage-energy'. Finally, the 'flux' tally was used to measure the neutron flux through the blanket. OpenMC calculates flux by tracking neutron paths in units of neutron-cm per source neutron. Integrating over the volume provides us with a value for the fluence (neutrons cm^{-2}).

3.3 Machine Learning Implementation

To gain a full understanding of the blanket behaviour it became necessary to utilise a machine learning framework. For this we used BoTorch (Balandat et al., 2019), a machine learning algorithm specialising in Bayesian optimisation.



(a) 2D Hidden Peak Test (b) 2D Schwefel Test

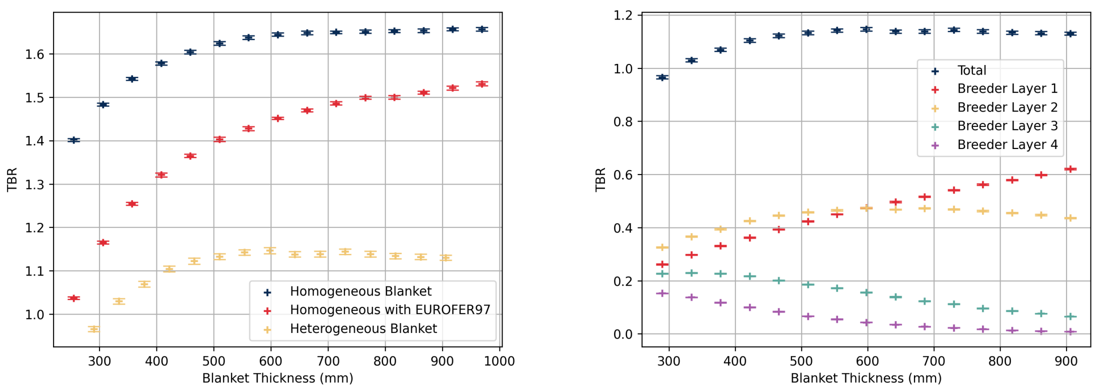
Figure 3: Two graphs demonstrating the results of 2D tests to validate the functionality of our machine learning model. Figure a shows the results for a hidden peak test while figure b does the same for a Schwefel function. Each figure contains a plot of the parameter space and explored datapoints with the mean and uncertainty of the surrogate function shown below.

We began by carrying out separate 4-dimensional investigations, optimising TBR for the thicknesses of the four breeder layers and then the four coolant layers. Then we carried out an 8-dimensional simulation of all layer thicknesses. In all cases the workflow was essentially the same. A set of 50 TBR values were generated using random points in the parameter space. Gaussian Process Regression was used to create a surrogate model using these values. An acquisition function defined by the UCB method (equation 7) was then used to select a configuration to explore. A value of β=1,000,000 was selected to ensure that the model prioritised the exploration of areas with high uncertainty in the parameter space. The blanket defined by these parameters was simulated and the TBR added to the dataset, modifying the surrogate model. To prevent the algorithm from becoming fixated on a narrow parameter space a random injection parameter (ϵ=0.5) was also implemented. The acquisition process was repeated 50 times to evaluate a blanket configuration that resulted in optimal tritium breeding.

To guarantee the validity of our machine learning programme it was necessary to carry out a series of low dimensionality tests prior to full implementation. The results of two example tests are shown in figure 3. Figure 3a depicts the results for a 2D hidden peak test. A short, wide peak and a tall, narrow peak are present which tests the algorithm's performance when exploring a narrow maxima. Meanwhile, figure 3b outlines the results of a Schwefel function test (Schwefel, 1981), where there are several similar peaks spread throughout the grid. This function aims to evaluate the programmes ability to comprehensively explore the parameter space when there are many candidate peaks. In both examples the machine learning implementation isolated the true peak and produced an acceptable surrogate model, as seen beneath the main plots. These tests confirmed the functionality of the machine learning framework and allowed us to proceed to higher dimensionality simulations with confidence.

4 Results

4.1 Heterogeneous Blanket Results



(a) Homogeneous - heterogeneous comparison (b) TBR of breeder layers

Figure 4: Graphs demonstrating the effects of blanket thickness on the TBR. Homogeneous and heterogeneous blankets are compared in figure a. Figure b shows the tritium breeding contributions of each layer in the heterogeneous blanket.

We began with a straightforward investigation into the effect of introducing heterogeneity into a blanket on its tritium breeding capabilities. TBR was recorded at a range of blanket thicknesses for homogeneous and heterogeneous configurations of a water cooled, $Li_{17}Pb_{83}$ breeder. The breeder and coolant layers were increased proportionally so that the relative size and coolant fraction remained constant. Separator layers in the heterogeneous blanket were kept at the same thickness throughout. A simulation was also carried out for a homogeneous blanket with a fraction of Eurofer97 proportional to that found in the separator layers of the heterogeneous blanket. Uncertainties of $\sim$0.5% were measured for all tallies.

As shown in figure 4a, the heterogeneous design has a significantly lower TBR than both homogeneous designs, not very much higher than equilibrium. Figure 4b depicts the contribution of each breeder layer to the total TBR at various total thicknesses. Breeder layer 1 was situated closest to the neutron source with breeder layer 4 furthest away. The fore layers yielded the highest TBR with layer 2 generating the most in thin blankets and layer 1 producing more with increasing thickness.

4.2 Plasma Body Neutron Source

In addition to modifying the blanket structure, the ring neutron source used in the previous semester was replaced with more realistic plasma body source. The parameters used to define the plasma body are specified in Fausser et al., 2012. TBR is plotted against triangularity for a heterogeneous blanket employing both neutron sources in figure 5. Both sources resulted in a virtually identical TBR with significant overlap of error bars. Neither source demonstrated a discernible trend between triangularity and tritium breeding.

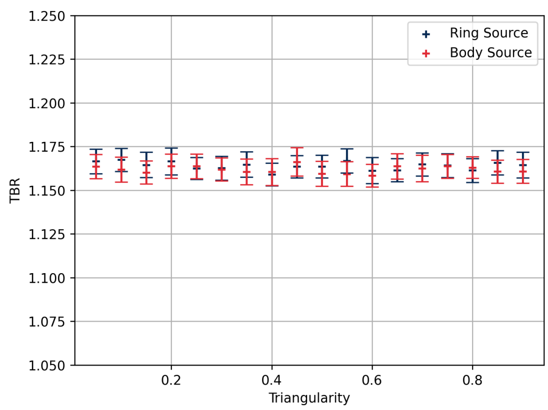
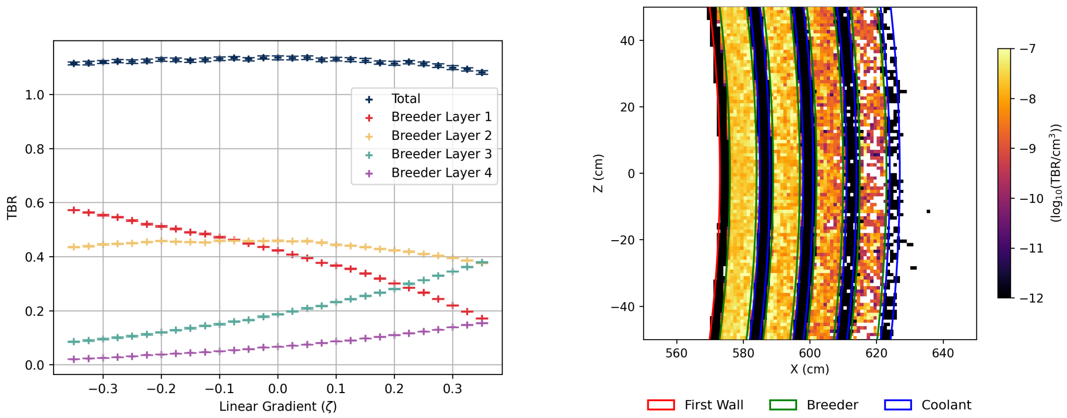


Figure 5: The TBR at various triangularity values for a heterogeneous blanket using a ring neutron source and a plasma body neutron source.

4.3 Linear Gradient Results

4.3.1 TBR

Once it was established that heterogeneity resulted in a noticable reduction in TBR it became necessary to probe this behaviour and deepen our understanding of the breeder behaviour. The linear gradient parameter ζ proved useful in this regard, allowing for the investigation of effects in the front and rear of the blanket. It is worth noting that all tallies are given in units per source neutron. Figure 6a exhibits the TBR contribution from different breeder layers at different ζ values. For more front-loaded designs (negative values) the first layer dominates, however at ζ>0 the second layer produces more tritium. Layers 3 and 4 contribute comparatively little, except at large, positive ζ. On the right, figure 6b shows the tritium breeding reactions taking place throughout the structure for ζ=0. As we would expect almost all tritium breeding reactions occur in the breeder layers, decreasing with depth into the blanket.

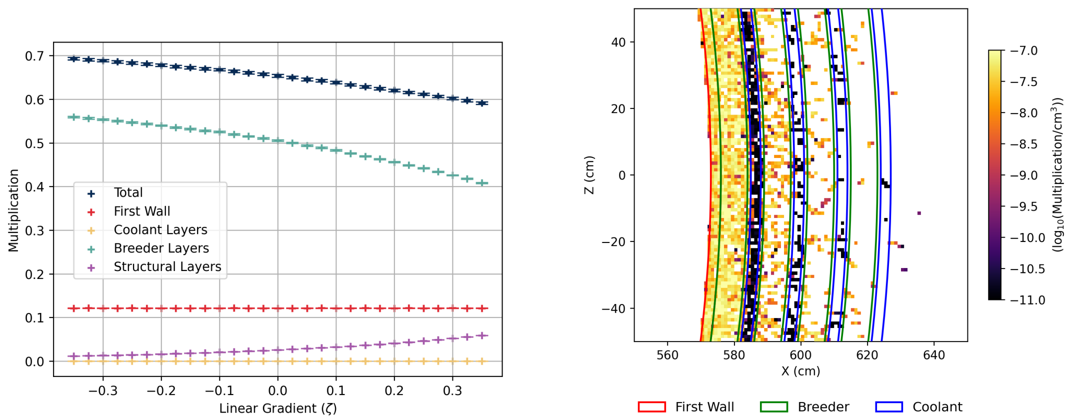


(a) Breeder TBR contribution (b) Tritium breeding reaction density

Figure 6: This figure contains two graphs. In figure a, the TBR contribution of each breeder layer is plotted for a range of ζ values. Tritium breeding reaction density is shown for a 2D slice of the breeder in figure b.

4.3.2 Multiplication

As previously stated, neutron multiplication reactions (n,2n) are necessary to ensure a TBR greater than equilibrium. The graphs in figure 7 depict the contribution from each layer type on the left and a heatmap of multiplication density on the right.



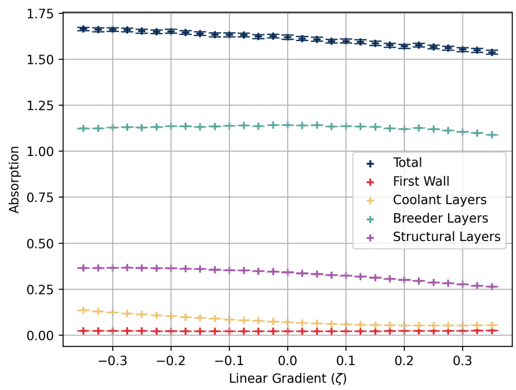
(a) Neutron multiplication (b) Multiplication reaction density

Figure 7: The graph on the left (figure a) displays the neutron multiplication produced in the blanket at different values of ζ. On the right (figure b) a 2D visualisation of the multiplication density is shown.

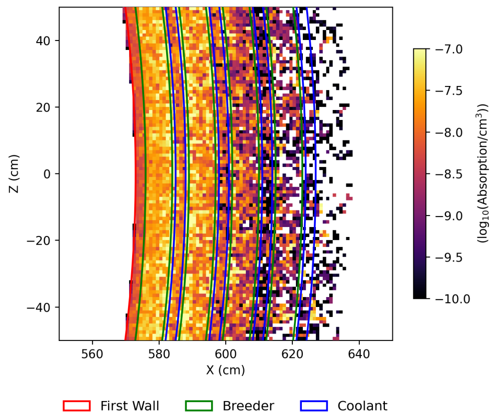
The majority of multiplication occurs in the breeder layers where neutrons interact with lead nuclei (equation 4). However, there is a small contribution from the zirconium in the first wall and the iron in the separators. With increasing ζ the multiplication in the breeder layers and the blanket overall declines steadily, despite a small increase within the structural layers (figure 7a). Almost the entirety of the multiplication takes place in first and second layers, as shown in figure 7b.

4.3.3 Absorption

To maintain a positive neutron economy, it is vital that parasitic neutron absorption is kept to a minimum. As seen in figure 8a, the absorption rate of each layer is plotted against ζ where the total declines steadily. The heat map in figure b shows the absorption reaction density, decreasing throughout the depth of the blanket. Since the absorption tally also counts tritium breeding reactions, the breeder layer has the highest recorded values, followed by the Eurofer97 separators.



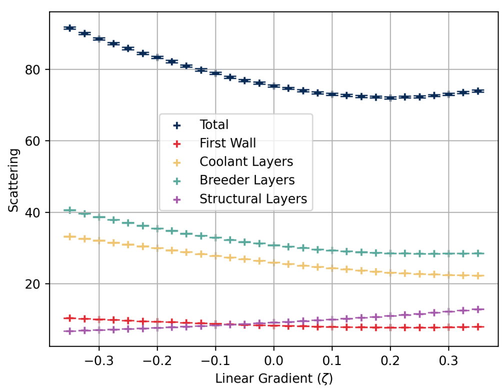
(a) Absorption contribution



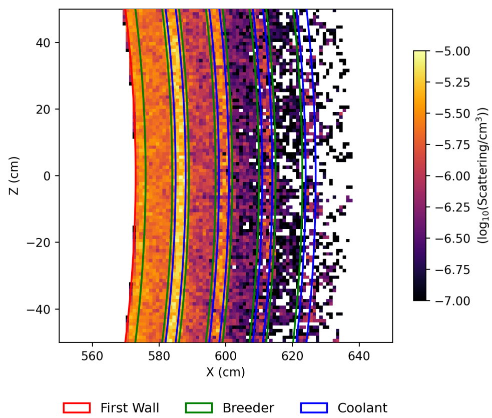
(b) Absorption reaction density

Figure 8: Two graphs showing the absorption measured in the breeder blanket. A graph of absorption at several ζ values is shown in figure a with a 2D heat map of the absorption in figure b.

4.3.4 Scattering



(a) Scattering contribution



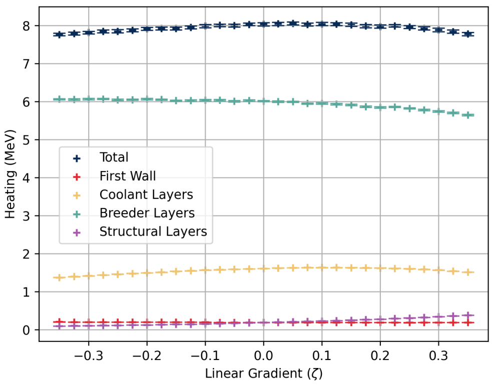
(b) Scattering reaction density

Figure 9: These figures demonstrate the change in scattering with values of ζ (a) and the scattering density in a 2D slice (b).

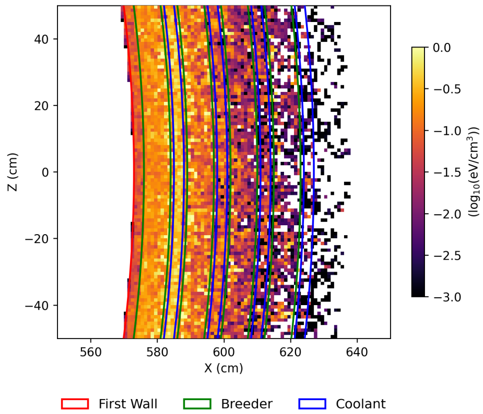
In addition to absorption, neutrons can also be scattered when interacting with nuclei, often losing energy in the process. Since a single neutron can be scattered many times the reaction rates recorded in figure 9a are much higher than for previous tallies. As ζ increases the total scattering declines before rising slightly. Scattering primarily takes place in the breeder as a result of its high density and in the coolant layers due to the high scattering cross section of hydrogen. This can be seen clearly in figure 9b.

4.3.5 Neutron Heating

Upon absorption or scattering neutrons may impart energy into the nuclei, heating the surrounding material. The leftmost graph of figure 10 demonstrates that heating energy remains roughly stable over ζ. Heating energy is overwhelmingly produced in the breeder layers with only a small contribution from the coolant layers and very little else besides.



(a) Heating energy contribution (MeV)

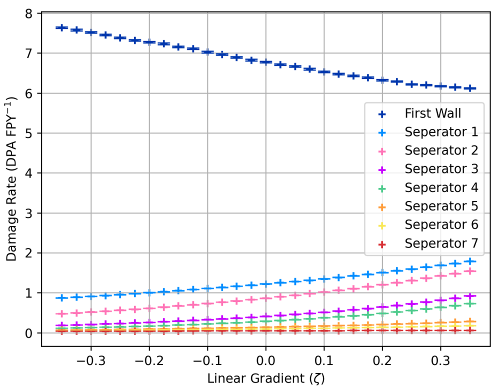


(b) Heating energy density

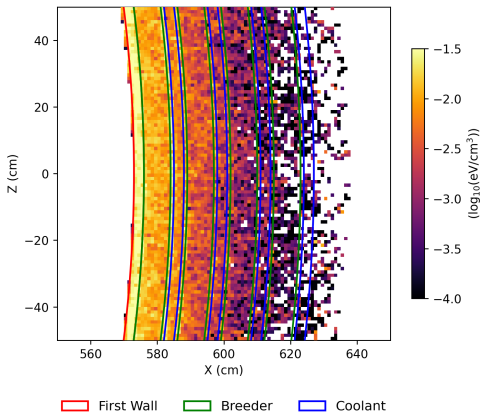
Figure 10: The heating energy (MeV) deposited in the blanket structures at values of ζ is depicted in a. The heating energy density in a 2D slice of the breeder is shown in b.

4.3.6 Damage Energy

OpenMC also allows the damage energy imparted to a material to be recorded.



(a) dpa fpy^{-1}



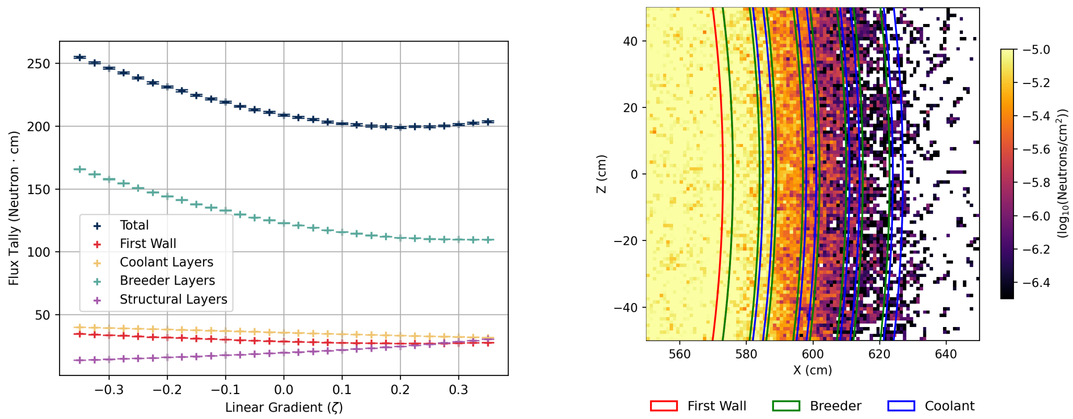
(b) Damage energy density

Figure 11: Figure a plots the dpa fpy^{-1} for a range of ζ values while the damage energy deposited in the blanket regions is visualised in figure b.

Equations 5 and 6 can be used to convert damage/displacement energy into an expression of displacements per atom per full-power-year (dpa fpy^{-1}). An assumption of a reactor power of 500MW was made based on the planned output of ITER (Aymar, Barabaschi and Shimomura, 2002). Since both the coolant and breeder are constantly circulating liquids it is only necessary to investigate the damage to the first wall and structural layers. A graph plotting the dpa fpy^{-1} against ζ is depicted in figure 11a, with a heat map of damage energy density in figure 11b. As we might expect the first wall is subjected to a displacement rate much higher than the rest of the reactor structure however, this value declines for rear-loaded (high ζ) configurations.

4.3.7 Neutron Flux

Flux is calculated in OpenMC by tracking the neutron paths through a volume and returning a measurement in neutron-cm. Dividing by the volume of the element provides a fluence (time integrated flux). As exhibited in figure 12a the breeder layers experience the largest integrated fluence due to their large volume. The heatmap in figure 12b demonstrates that the fluence decays exponentially at increasing blanket depth.



(a) Neutron track length

(b) Fluence (neutrons cm^{-2})

Figure 12: These figures demonstrate how neutron transport changes in the blanket. The effect of ζ on the neutron track length measured by the 'flux' tally is shown in figure a. Figure b demonstrates the exponential decay of fluence throughout a 2D slice of the breeder.

4.4 Machine Learning Results

In addition to the manual investigation of blanket behaviour a machine learning framework was used to optimise the breeder geometry. The thicknesses of each breeder and coolant layer served as variables that would be explored to evaluate a peak TBR. The breeder layers were limited to between 30 mm and 160 mm and the coolant layers between 15 mm and 60 mm. Optimum configurations and the resulting TBR values are displayed in figure 13. Breeder layers are labelled B1 - B4 and coolant layers C1 - C4, counting radially outwards.

To begin with, 4D simulations were carried out where only the breeder layers or only coolant layers were allowed to vary, leading to TBR values of 1.2805 ± 0.0026 and 1.2080 ± 0.0011 respectively. When varying the breeder layers, their total thickness was capped at 320 mm while the coolant sizes were fixed at 150 mm. Meanwhile in the coolant only simulation their collective thickness was set to 120 mm with 80 mm breeder layers.

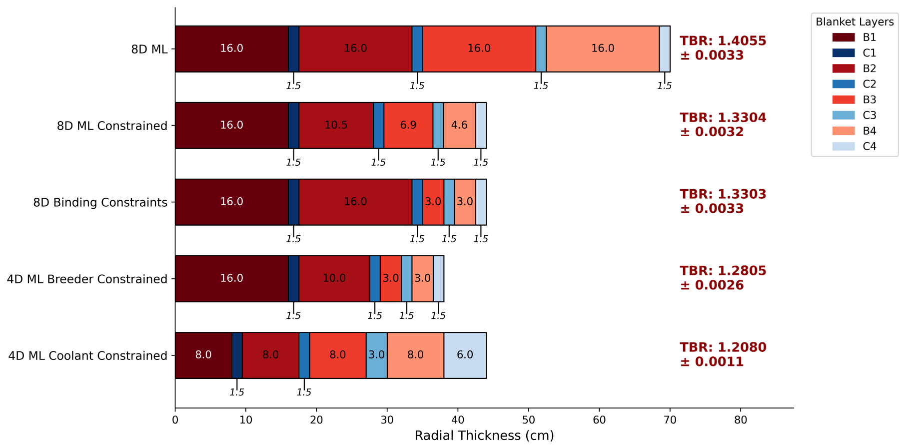
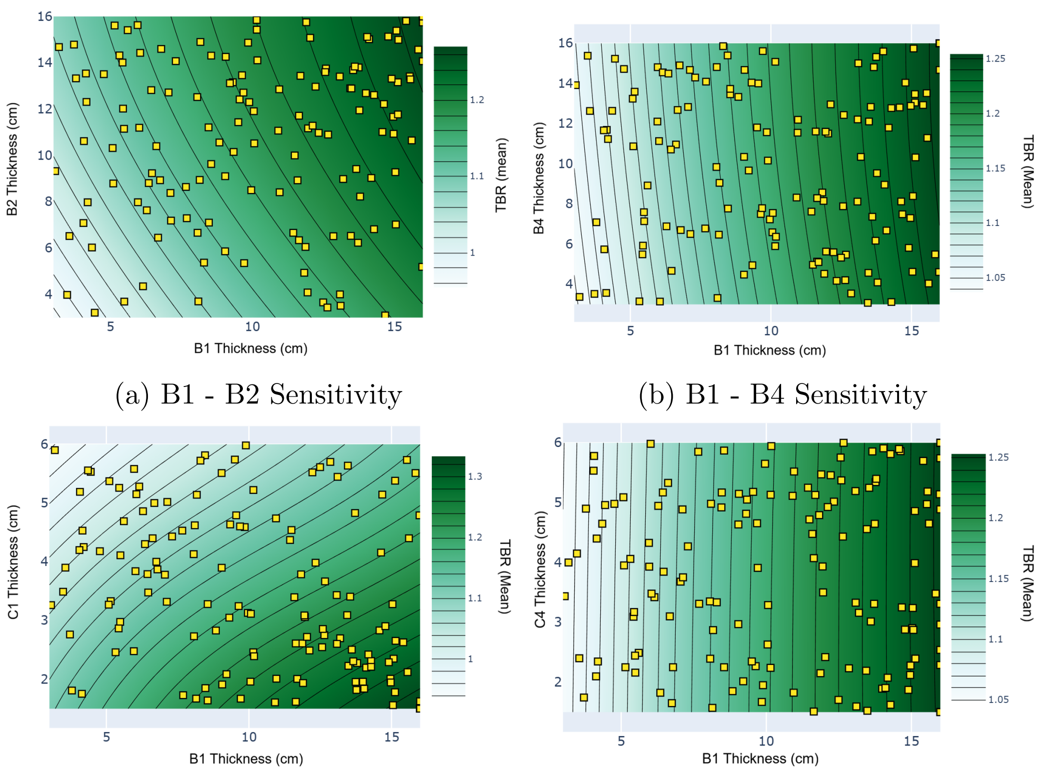


Figure 13: A graph that exhibits the optimum blanket configurations for several 4D and 8D machine learning simulations and their resulting TBR.

Once completed, a series of 8-dimensional investigations were performed where all layers could be modified. A constrained 8D investigations was performed, limited to a total thickness of 440 mm. This resulted in a configuration of exponentially decaying breeder thicknesses and minimised coolant layers with a TBR of 1.3304 ± 0.0032. A virtually identical result of 1.3303 ± 0.0033 was evaluated with a manually defined 'binding constraint' that front-loaded the breeder material. The final simulation removed all constraints on thickness so as to produce a maximum TBR of 1.4055 ± 0.0033.



(a) B1 - B2 Sensitivity

(b) B1 - B4 Sensitivity

(c) B1 - C1 Sensitivity

(d) B1 - C4 Sensitivity

Figure 14: 2D projections of the sensitivity of TBR to variations in the parameter space. The relative influence of various layers (B2, C1, B4 and C4) is compared with that of the first breeder (B1). Straight vertical lines indicate very little effect relative to B1. Curved lines suggest a similary strong response from both variables.

To evaluate the relative contributions of each layer to the 8D machine learning result a series of plots was created visualising the interaction between two parameters in the simulation. In figure 14 the first parameter sensitivity of the first breeder layer (B1) is compared with layers B2, B4, C1 and C4. Straight vertical lines indicate that the contribution of the layer under investigation is relatively inconsequential compared to B1. On the other hand, curved lines suggest that both parameters have material effect on the TBR. The figures confirm that TBR is highly sensitive to variations the layers closer to the front (B2 and C1) while the rear layers (B4 and particularly C4) have negligible effects.

5 Discussion

5.1 Heterogeneity and Blanket Thickness

The data exhibited in figure 4a clearly demonstrates that a heterogeneous blanket results in a significantly reduced TBR compared with a homogeneous blanket with a comparable chemical composition. In a homogeneous breeder the water coolant acts as a moderator, reducing neutron energies to values where the tritium production cross section increases by several orders of magnitude (Brown et al., 2018). Clearly in a heterogeneous configuration this is not the case. Neutrons are already moderated by scattering within the first breeder layer and are much more likely to be absorbed by the separators or the coolant. The TBR of both homogeneous designs begins to plateau at a total thickness of ∼500 mm, however they still increase slowly beyond this. Since neutron transport is probabilistic there will still be a very small increase in TBR even at very large thicknesses. Meanwhile as the depth of the heterogeneous blanket increases beyond 500 mm the TBR completely levels and even begins to decline slightly. A homogeneous blanket with a proportional fraction of Eurofer97 to the separator layers of the heterogeneous design was also studied to confirm that the reduction of TBR was due to blanket layering. The introduction of Eurofer97 leads to some reduction in TBR, but it is clear that heterogeneity has a distinct effect. Most interestingly, the rate of TBR increases in the homogeneous Eurofer97 design above 500 mm and continues to rise at a faster rate than the homogeneous design without Eurofer97. By adding Eurofer97 the overall blanket density is significantly reduced, leading to deeper neutron penetration and tritium breeding.

The layered nature of the breeder means that with greater thickness the coolant layer between the first two breeders increases, leading to greater absorption and a reduction in the tritium production of layers 2, 3 and 4. This can be seen clearly in figure 4b. With greater thickness the contribution of breeder layer 1 increases linearly. Meanwhile, that of the second layer peaks at ∼500 mm before declining steadily. Tritium breeding in layers 3 and 4 decline consistently with total thickness. This behaviour is to be expected as thicker front layers reduce neutron transport to the rear of the blanket. For thinner blankets the TBR of layer 2 is in fact higher than that of the first layer. Neutrons that reach the second layer have lower energies having been scattered by the first. Essentially, the neutrons interacting with layer 2 have been moderated. As neutron energy declines the tritium breeding cross section of ^{6}Li increases by several orders of magnitude (Brown et al., 2018), driving up the TBR. However, as the depth of layer 1 increases so does its tritium production capacity while reducing quantity of neutrons reaching the second layer.

5.2 Neutron Source

Adopting a plasma body neutron source as a replacement for the ring source had a negligible effect on the resulting TBR, demonstrated in figure 5. TBR also remained uncorrelated with the reactor vessel triangularity. We had expected that a more spread out body source might exhibit some variation as a result of changes to neutron flux distribution. However, the TBR remains stable. These results suggest that the selection of breeder triangularity can be made to match material and structural requirements without significantly affecting its tritium breeding capability.

5.3 Linear Thickness Gradient

5.3.1 Tritium Breeding Ratio

A linear gradient was imposed on the blanket layers using the factor ζ, allowing for the investigation of neutronic behaviour in front-loaded and rear-loaded blankets. Tritium breeding plotted in figure 6a remained relatively constant at $\geq$1.05 over the range of ζ values. As we would expect, breeder layer 1 contributed the largest TBR in more front-loaded configurations but declined steadily with increasing ζ. TBR in layer 2 remained relatively constant across ζ values while it increased steadily in layers 3 and 4. At identical breeder volumes (ζ=0) the second layer TBR is higher than the first. This can be understood by considering moderation. Neutrons absorbed in the first layer have undergone less scattering than those incident on layer 2 and thus have higher energies. This results in a lower tritium production cross section (Brown et al., 2018) and a reduced TBR for the same volume. A visualisation of TBR reaction density can be seen in figure 6b for ζ=0. Once again, the TBR of the first two layers is shown to be much higher than the rear two. Outside of the breeder layers there is a small but ultimately negligible contribution to TBR due to tritium production reactions involving zirconium in the first wall and deuterium in the coolant. The presence of these reactions was discussed in the first semester report (Peters, 2025) and can be ignored due to their low reaction cross sections (Brown et al., 2018). Overall, the tritium breeding behaviour is in line with our understanding of neutron transport, reflecting the changes in the breeder volume in each layer along with greater neutron penetration into the rear layers at higher values of ζ.

5.3.2 Neutron Multiplication

Neutron multiplication is particularly biased towards the front layers, even more so than TBR. This is clearly visible in figure 7a where the multiplication declines consistently with increasing ζ and in figure 7b where the density of multiplication reactions is almost negligible beyond layer two. Tritium breeding is not only possible but is more likely at thermal neutron energies. The same cannot be said for the neutron multiplication reactions of lead isotopes where there is a $\sim$7.5 MeV energy threshold (Brown et al., 2018). Thus, the rate of neutron multiplication declines significantly beyond the second breeder layer as they lose their energy in scattering reactions within the first breeder and coolant layers. A contribution to the total is also recorded by the zirconium in the first wall. This value remains constant across ζ since the reaction has a threshold energy of $\sim$12 Mev. While many neutrons produced by multiplication in the first breeder layer are scattered back towards the first wall, they do not have sufficient energy to incite multiplication in the zirconium. In short, multiplication in the first wall is solely the product of direct source neutrons, thus the value remains constant as blanket geometry is varied. Meanwhile the (n,2n) reactions that take place in the reactor layers by interacting with iron nuclei gradually increase as the blanket becomes more rear-loaded. A thinner first breeder layer results in a greater number of more energetic neutrons interacting with the separators, increasing the multiplication rate.

5.3.3 Neutron Absorption

There are two neutron absorption processes of note that take place in a breeder blanket; the absorption of neutrons that results in tritium production, and parasitic absorption. Breeder blanket design is aimed at increasing the former and keeping the latter to a minimum. As depicted in both figure 8a and 8b, the largest contribution to absorption takes place in the breeder layers as neutrons are absorbed to release tritium. Consequently, this value remains relatively constant over ζ, matching the TBR trend demonstrated in figure 6a. The largest contribution to absorption outside of the breeder material occurs in the Eurofer97 separators. This parasitic absorption decreases with increasing ζ. Thinner first layers result in greater neutrons travelling further and interacting with the separator material at higher energies. Iron isotopes have lower absorption cross sections at higher incident neutron energies (Brown et al., 2018), leading to the decrease in absorption. Very little absorption takes place in the first wall of the breeder despite its proximity to the neutron source. This is due to the relative neutronic transparency of zirconium. Thus, it is no surprise that zirconium is a particular favourite among first wall candidate materials.

5.3.4 Neutron Scattering

Neutrons travelling through the blanket structure can be scattered off nuclei several times before they are absorbed. As ζ increases in figure 9a, the total scattering rate declines before plateauing and gradually increasing at $\zeta>0.2$. This is likely due to the decline in multiplication and the commensurate decrease in neutron population. The contributions to the total from the coolant and breeder layers are relatively similar despite the much greater volume of the breeder. Figure 9b shows the much higher scattering density that takes place in the first coolant layer, highlighting the superior moderation properties of water. The small increase in scattering at high values of ζ is a result of the separator layers. Thinner front layers result in greater penetration and higher energy neutrons interacting with the separator. In contrast to its absorption cross sections, the scattering cross section of iron isotopes is much larger at higher energies. This accounts for the increase in scattering.

5.3.5 Heating Energy

Incident neutrons release energy into a material during inelastic scattering and certain absorption reactions. The total heating displayed in figure 10a remained relatively constant at around 8 MeV per source neutron. Most heating occurred in the breeder layers as a combined result of a high scattering rate and the energy released during the exothermic Li_6 tritium breeding reaction (equation 2). The high scattering cross section and low mass of the hydrogen in the water of the coolant layer allows it to extract a significant amount of energy from incident neutrons, resulting in the next highest contribution to the overall heating. In general, it is preferable that these materials, particularly the coolant, be subject to the majority of heat production since they are liquids that can be circulated to distribute heat more effectively.

5.3.6 Damage Energy and Lattice Displacement

In addition to heating the material, energy deposited by neutrons can displace atoms from the crystal lattice. Using equations 5 and 6 the dpa imparted to the first wall and each separator in a full-power-year is calculated and recorded in figure 11a. Since the breeder and coolant are both liquids and not vital to the structural integrity of the blanket they were ignored during this section. As can be seen in figure 11a and 11b the first wall receives the vast majority of damage energy and resulting dpa. This is as we would expect due to its proximity to the neutron source. The dpa fpy^{-1} received by the first wall decreases from 7.52$\pm$0.02 to 6.18$\pm$0.01 across the range of ζ values. Higher ζ leads to less neutron multiplication in the first layer and less neutrons scattered back into the first wall and less damage. On the other hand, higher ζ increases neutron penetration separator layer damage.

Material irradiation can result in a wide variety of processes that have complex effects on the properties of reactor components. Helium induced embrittlement resulting from transmutation reactions is the method by which we will characterise material failure. Gilbert et al. (2012) provides dpa thresholds for a variety of materials and grain sizes. For the zirconium in the zircaloy-4 first wall with a grain size of 5 μm (Woods, 1966) the critical helium embrittlement occurs at a dpa of 61.99. This would suggest a first wall lifetime of 8.23$\pm$0.02 to 10.03$\pm$0.02 full power years, significantly lower than the 30 years planned for ITER (Aymar, Barabaschi and Shimomura, 2002). While zirconium based materials are neutronically superior they are clearly inadequate from a materials stability standpoint. Performing a similar investigation with a Eurofer97 first wall resulted in a $\sim$60% higher absorbed damage energy but a significantly lower dpa fpy^{-1} of between 2.18$\pm$0.01 and 2.51$\pm$0.01. Stornelli et al., 2024 describes Eurofer97 a with grain size of 0.5 μm. A threshold dpa of 57.47 results in a full-power lifetime of 22.90$\pm$0.05 to 26.41$\pm$0.06, much closer to what would be required of a first wall in a functional fusion reactor. Unfortunately, the Eurofer97 first wall reduces the TBR by 9%, below the threshold of 1.05 required for reliable tritium breeding. This underlines a pressing issue in fusion research, where material properties place severe constraints on design and compromises on neutronic performance may be necessary to guarantee safety.

5.3.7 Neutron Flux

Flux is measured by tracking neutron paths through the breeder structures. Figure 12a shows that the total neutron track length calculated by OpenMC's 'flux' tally declines as ζ increases before reaching a plateau at ζ=0.2. This is likely a result of the reduction in multiplication rate demonstrated in figure 7a. A lower neutron population leads to less neutron travel per source neutron. The breeder layers serve as the main contributor to the total flux as the largest elements by volume and the source of most neutrons born from multiplication reactions. Breeder, coolant and first wall flux all decline steadily across the range of ζ values. However, the neutron track length measured in the separator layers increases linearly as the first breeder layer becomes thinner, allowing greater neutron penetration into the structure. This confirms the scattering and damage energy behaviour discussed previously, where greater neutron transmission depth and incident energy led to increases in these values. The fluence in a 2D segment of the breeder is visualised in figure 12b. As expected, the fluence decays exponentially with depth into the blanket, leading to the significantly higher neutronic contribution of layers 1 and 2 compared with layers 3 and 4.

5.4 Machine Learning and TBR Optimisation

As could be expected the 8-dimensional, unconstrained algorithm resulted in the highest TBR of the trialled machine learning models with a value of 1.4055±0.0033. Breeder thicknesses were maximised at 160 mm and coolant layers minimised at 15 mm. While this undoubtedly results in a higher TBR a blanket of this size would be inefficient and costly to implement. In an attempt to remedy this issue the total blanket thickness (not including separator layers) was limited to 440 mm. With this constraint the machine learning model produced a decaying configuration with minimised coolant thicknesses and breeder thicknesses of 160 mm, 105 mm, 69 mm and 46 mm. The decreasing breeder thickness led to a TBR of 1.3304±0.0032. This configuration and the resulting TBR are remarkably similar to the 'binding' blanket design which selects the maximum value for the first two breeder layers and the minimum value for the rear layers. These results demonstrate clearly that the first two layers are prioritised in terms of tritium breeding, which is in agreement with our results using the linear gradient factor (ζ). Additionally, while the presence of water increased the TBR in a homogeneous breeder, the machine learning algorithm prioritises the reduction of coolant volume.

When limited to 4-dimensional investigations the peak TBR is appreciably lower. Allowing only either the breeder layers or the coolant layers to vary resulted in tritium breeding ratios of 1.2805±0.0026 and 1.2080±0.0011 respectively. Much like the constrained 8D simulations the 4D breeder model recreated the frontloading phenomena. Conversely, the 4D coolant simulation produced a configuration with coolant layers increasing in thickness further into the blanket. Once again, we see that the tritium breeding effects are concentrated in the first two layers. When limited to a set total depth the model prioritises these layers above the rear two. These insights may be useful when designing a blanket that requires a specific overall coolant flow or has a maximum total depth.

The parameter sensitivity graphs exhibited in figure 14 were used to validate the relationships between individual layers observed in the 8-dimensional investigations. The contribution to the TBR of layers B2, B4, C1 and C4 were plotted against B1. Curved contours shown in figures 14a and 14c confirm that layers B2 and C1 were significant contributors to the TBR optimisation. Meanwhile, the near straight lines of figures 14b and 14d show that the rear layers of B4 and C4 were relatively unimportant to the machine learning programme. They confirm the trends inferred from figure 13, that the variation in the TBR is primarily a result of modifications to the thickness of the first two layers.

5.5 Limitations and Potential Improvements

Our investigation into the tritium production of heterogeneous breeder blanket has resulted in the modelling of optimised breeder designs and deepened our understanding of the neutronics involved. However, with only a single semester in which to explore a layered blanket configuration, the scope of our investigation was incredibly limited. The selection of a single breeder and coolant material was made based on this constraint. If further study were to be carried out a comprehensive exploration of blanket materials would certainly merit significant attention. Especially since the chosen composition was based on the best performing materials in a homogeneous breeder. As has been demonstrated, heterogeneous blanket neutronics result in very different tritium breeding behaviour.

A blanket with four layers each of alternating breeder and coolant was selected as compromise based on the wide variability in the available literature. More complete research would aim to confirm whether this was indeed the optimal tritium breeding configuration. The number of layers and the ordering of breeder and coolant would have significant effects on neutron transport through the blanket. More complex blanket structures such as piping, containment and even the pebbles in a solid breeder can be modelled, although with significant difficulty. Our modifications to the original homogeneous design have provided interesting results, however many improvements would be required to model a realistic breeder blanket.

The Bayesian optimisation machine learning programme implemented to explore the TBR behaviour of the blanket successfully identified the optimal breeding configuration for a series of 4D and 8D investigations. Despite its clear advantages when exploring the behaviour of a multidimensional parameter space, improvements could still be made. The successes encountered at a series of lower parameter investigations suggest that the machine learning algorithm is scalable to higher dimensions, allowing for the optimisation of more complex blankets as discussed above. Multi-objective optimisation, where the values several tallies are considered when varying the parameter space, is a particularly interesting avenue of exploration that would ensure that blankets were evaluated on several fronts, not just TBR.

6 Conclusion

Using a dedicated neutron transport Monte Carlo simulation we have explored the TBR and neutronic behaviour of a heterogeneous, water cooled, lithium-lead breeder blanket. As we had predicted beforehand, the introduction of heterogeneity led to diverging neutronic properties, resulting in a significantly lower TBR than an equivalent homogeneous configuration. Our results also emphasised the importance of considering factors beyond tritium production evaluating blanket designs. In particular, the irradiation damage absorbed by the first wall was far greater than would be acceptable in a practical breeder blanket. While zirconium alloys offer a superior neutronic performance they lack the structural properties required to withstand the harsh environments of operational fusion reactors.

Implementing a machine learning workflow allowed TBR to be optimised for breeder and coolant layer thicknesses. Unconstrained optimisation led to maximised breeder and minimised coolant sizes in agreement with predicted neutronic behaviour. Meanwhile, enforcing a total thickness limit produced a front-loaded design with breeder thickness decaying exponentially away from the first wall. Clearly, the breeder layers closer to the source are of greater importance to tritium production than the rear layers. The machine learning algorithm succeeded in identifying much higher TBR values than our manual investigation and replicated the trends expected from our understanding of neutron behaviour. These results demonstrated the utility of simple machine learning programmes when carrying out optimisation processes in multivariable parameter spaces.

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